The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AP30023
Subject Title	Designing Sensing Systems for Internet of Things in Smart Cities
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	
Objectives	1. To provide fundamental knowledge on physical principles of sensors and transducers for measuring essential physical signals (temperature, humidity, light, position, velocity, acceleration, pressure, air quality, water quality, electric current and so on) needed in Internet of Things. 2. To provide fundamental knowledge on application of digital processing techniques to improve sensor data accuracy or extract insights from big data or multiple data sources. 3. To provide a comprehensive understanding on the requirements of the design, calibration and operation of sensing systems for Internet of Things. 4. To provide training and impart a reasonable level of competence in the design of sensing systems for Internet of Things applications in Smart Cities.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Understand the architecture of Internet of Things and the typical power, communication and signal interfacing requirements of its sensing system. b. Understand the principle of sensors and transducers for converting physical parameters into electrical signals. c. Evaluate the performance of different types of sensors based on figures of merit. d. Identify the applications of different types of sensors for Internet of Things. e. Design and develop sensing system by selecting suitable sensors/transducers to achieve desirable properties based on application requirements and system architecture. f. Compete in the design, construction, and operation of sensing systems for measuring different physical quantities.
Subject Synopsis/ Indicative Syllabus	1. Fundamentals of Internet of Things architecture - Network architecture, system elements, signal interfacing, power requirement and energy harvesting 2. Fundamentals and characteristics of sensors

	- Sensor classification/type, principle, performance and error analysis. 3. Smart building* - Enhancing safety, security, comfort and efficiency 4. Smart parking & mobility* - Detection of object presence and driving pattern 5. Smart healthcare* - Detection of vital signs and home accidents 6. Smart agriculture* - Detection of air quality, water quality and plant growth 7. Implementation and operation issues in practice - On robustness, reliability, accuracy, interference and noise **Relevant sensors and digital processing will be introduced. Example list of sensors:* *i. Temperature sensors* *ii. Light detectors* *iii. Humidity and moisture sensors* *iv. Presence, displacement, level and position sensors* *v. Motion, velocity, acceleration and direction sensors* *vi. Force and strain sensors* *vii. Pressure and flow sensors* *viii. Chemical and biological sensors*
Teaching/Learning Methodology	The methodology includes classroom teaching, laboratory experiments and presentation. The principles and applications of different sensors for internet of things will be explained in lectures and tutorial, which are related to most of the learning outcomes (a–e). Lab sessions will be conducted to enhance the ability of the students in understanding and using various sensors (outcomes c and e). Students will give presentations on the design and applications of sensors and their findings (outcome f).
Assessment Methods in Alignment with Intended Learning Outcomes	(see table below)

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
		a	b	c	d	e	f
1. Continuous assessment	50	✓	✓	✓	✓	✓	✓
2. Course Project	50	✓	✓	✓	✓	✓	✓
Total	100 %						

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Continuous assessment is based on assignments, laboratory reports and test. The test is a one-hour written middle-term test. The course project is a sensor application design project which covers all of the key issues

	related to the learning outcomes. Students need to do project demonstration and presentation, and submit a project report.	
Student Study Effort Expected	Class contact:	
	▪ Lecture & Tutorial	32 Hrs.
	▪ Laboratory	9 Hrs.
	Other student study effort:	
	▪ Self-study	79 Hrs.
	Total student study effort	120 Hrs.
Reading List and References	*Handbook of Modern Sensors: Physics, Designs, and Applications*, Fraden, Jacob, 2016, 5th edition, Cham: Springer International Publishing. *Sensor Technology Handbook*, Jon S Wilson, 2005, Newnes. *Build your own IoT platform: develop a flexible and scalable internet of things platform*, Tamboli, 2022, 2nd edition. New York, NY: Apress. *IoT System Design : Project Based Approach*, James, Seth, Mukhopadhyay, 2022. Cham: Springer.	